RBE-3001 Final Project

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***Abstract*—Team 2's final RBE final project acted as a challenge to bring together all previous labs' topics and tasks. Incorporating aspects of forward kinematics, inverse kinematics, computer vision, and image processing, the project involved programming a robot arm to navigate about a space while identifying and physically sorting spheres of different colors. In addition to this primary task came the challenge to program dynamic object tracking, where a sphere of a given color would be consistently and smoothly followed by the robot's end effector, and a special challenge created by the team to have the robot play tic-tac-toe against a human. Over the course of the project, the most significant problems encountered and solved by the team consisted of identifying the source of software and programming difficulties.**

I. Introduction

A. Background

The team's work in Lab 5 would involve the combined use of the robot's end-effector, or gripper, which would be used to pick up and move different shapes, and its camera, which would be implemented so that the robot would be capable of identifying specific colors and shapes in a well-lit environment. With this being the case, the lab required the use of many previous concepts and code from earlier labs. Expanding upon these labs' material would consist of programming the robot's camera to identify and sort shapes based on their different physical forms and colors, with the arm being programmed in various different ways depending on how it was expected to treat differently colored items.

B. Motivation

The team's motivation to complete this lab was brought about by a general interest and skillset in robotics and kinematics that was developed prior to and over the course of RBE 3001. This motivation grew alongside the group's growing skillset in these areas, which allowed for the final project to serve as an illustrator of group members' abilities.

II. Methodology

The team's work to have the arm pick up and relocate the differently-colored balls began with the use of code from previous labs. Crucial to this goal was the code that had been written and edited over the course of labs 1, 2, 3, which would allow the robot to move to specific coordinates and along specific paths with the combined use of forward and inverse position kinematics. This would allow the arm to navigate about its environment in order to pick up balls at specific locations.

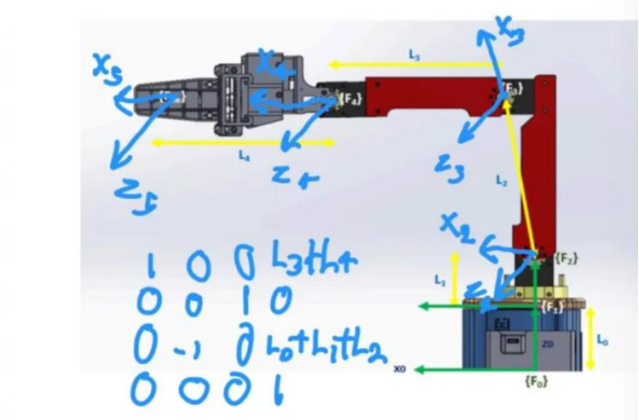


Figure 1: Reference Frames drawn on arm joints with final transformation matrix

T^{i}_{i-1}	θ_z (rad)	d_z (mm)	a_x (mm)	α_x (rad)
T^{1}_{0}	0	$L_0 = 36.076$	0	0
T^{2}_{1}	θ_2	$L_1 = 96.326 - 36.076 = 60.25$	0	$-\pi/2$
T^{3}_{2}	$\theta_3 - ((\pi/2)-\arctan(24/128)) = \theta_3 - 1.39$	0	$(24^2+128^2)^{0.5} = 130.23$	0
T^{4}_{3}	$\theta_4 + 1.39$	0	$L_3 = 124$	0
T^{5}_{4}	θ_5	0	$L_4 = 133.4$	0

Table 1: Denavit Hartenburg Table Corresponding to the Reference Frames Established in Figure 1

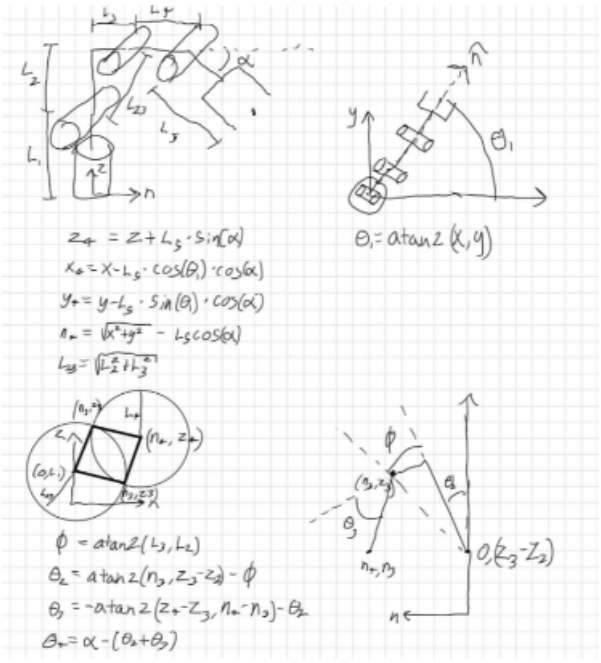


Figure 2: Inverse Kinematics derivation

Alongside the importance of position kinematics was the significance of forward and inverse velocity kinematics, which were implemented in the arm's code in lab 4 and would be necessary to ensure that balls could be picked up and moved around at the necessary speed without being dropped or otherwise mishandled.

In order to build upon the code developed during previous labs, the team began with the template code that was provided in the form of ImageProcessor.m and sort_balls.m. The functions crucial to these scripts would be detect_balls and pick_up_balls, which would be used in order to bring about the robot's recognition and retrieval of spheres of different colors, respectively.

Beginning the work that was unique to Lab 5 was the extrinsic and intrinsic calibration of the robot's camera. The first of these calibration types refers to calibration with respect to the camera's built-in, or intrinsic properties, and was done through the use of MATLAB's camera calibration app which allowed MATLAB to calculate intrinsic camera matrix. It was thought by the group that this was done by analyzing images of the known checkerboard pattern and using the relationships between pixel coordinates and real-world distances to estimate the camera's focal length, optical center, and distortion coefficients.

Following intrinsic calculation was extrinsic calibration, referring to the spatial calibration of the camera in relation to its surroundings. In this case, this extrinsic calibration consisted of using MATLAB's pointsToWorld() function to convert the camera's pixel coordinates to coordinates on the checkerboard between the camera and the rest of the robot.

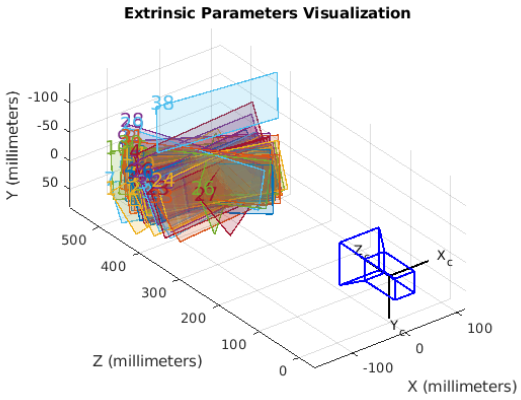


Figure 3: Extrinsic Parameters Visualization generated from camera calibration

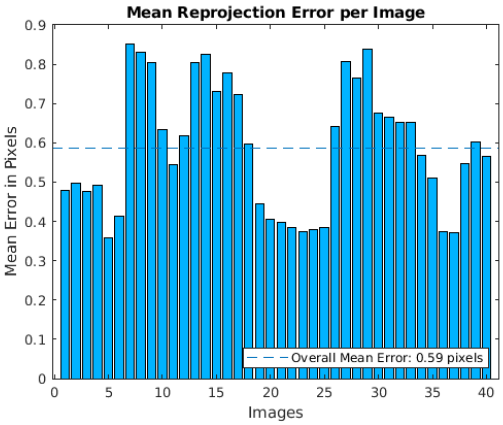


Figure 4: Mean Reprojection Error per Image generated from camera calibration

After calibrating the robot's camera, the camera's image was masked such that the areas outside of the checkerboard were ignored. Color masking was then implemented such as to ensure that all elements of the image excluding those of the spheres' colors were ignored. This was done using the Color Thresholder app in MATLAB's computer vision toolbox.

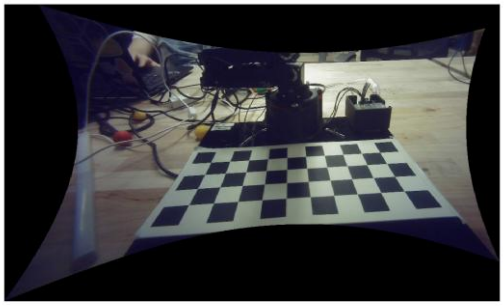


Figure 5: Correcting fish-eye distortion

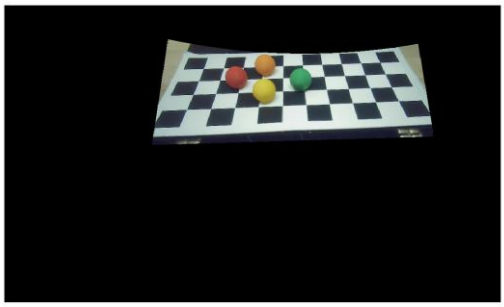


Figure 6: Correcting fish-eye distortion with objects and masking out background

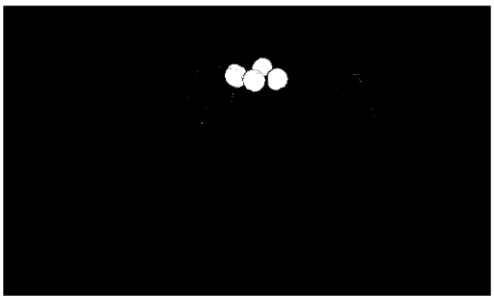


Figure 7: Masking out checkerboard to only see colored objects

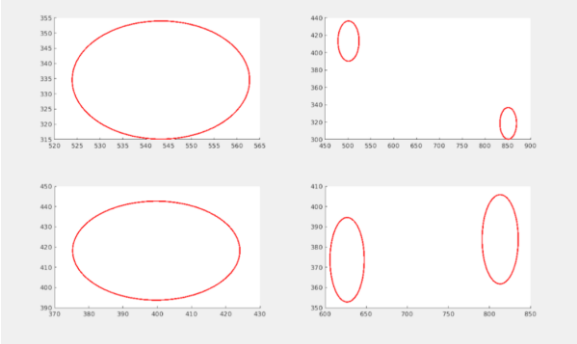


Figure 8: Circles found in the camera field and their positions

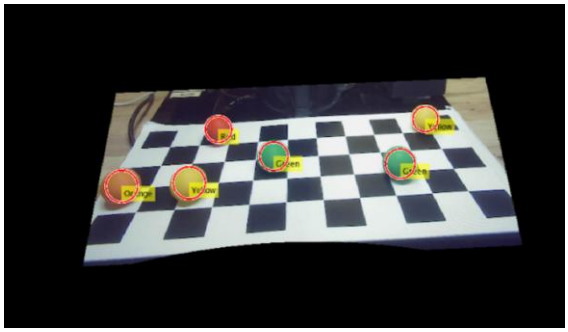


Figure 9: The circles labeled with their colors

After having programmed the robot and its camera to properly recognize the differently-colored spheres, the robot was programmed to respond to a sphere of a given color by, using the coordinates of its centroid, pick up the sphere and move it to preestablished coordinates. This was done with blocking code, as to use non-blocking code would allow for potential interruptions of the robot's movement by the camera's detection of new items. Pseudocode illustrating the robot's process to pick up a ball can be seen in Figure 10. The process shown in this pseudocode was chosen as the ideal manner of picking up a ball because it would ensure that the end effector would be further away from other spheres when moving a sphere from its initial coordinates to it's destination.

```
% sets the positions of each color's bins
switch color
            case "Red"
                BinPos = [50, 250, 150, 0];
            case "Orange"
                BinPos = [50, -250, 150, 0];
            case "Green"
                BinPos = [200, 250, 150, 0];
            case "Yellow"
                BinPos = [200, -250, 150, 0];
            otherwise
                BinPos = [100, 350, 150, 0];
        end

%open gripper
self.open_gripper();

%go above the ball's pos
self.blocking_ts_move([100, 100, 100, 90], 2);

%lower arm onto ball
self.blocking_ts_move([100, 100, 20, 90], 2)

%grab ball
self.close_gripper();

%lift the ball up
self.blocking_ts_move([100, 100, 100, 90], 2);

%move to bin
self.blocking_ts_move(inPos, 2);

%drop ball
self.open_gripper();
```

Figure 10: Pseudo Code to pickup and drop ball in bin

The first task to be performed by the robot was to sort spheres of different colors into four separate areas. This was done by having the camera take an initial reading of all of the spheres in front of it along with those spheres' coordinates. It would then proceed to pick up a sphere, move to preestablished coordinates, release the contents of its hand, and then repeat that process for the coordinates of each of its camera's initial readings.

Following the group's completion of the initial lab requirement, the robot was next programmed to implement real-time object tracking. This would allow the end effector to follow around a single sphere of any given color, with the color in question being written into code beforehand. At first this was very slow and we realized that our image masking was inefficient and was being slowed down by plot generation and print statements. After reducing those and making the masking more efficient we managed to get the arm to not lag behind from the balls real time position. We also added a filter so that the arm only follows a specified color and ignores objects of other colors.

The final task completed by the group consisted of programming the robot to play tic-tac-toe. This task was completed by first implementing the same object and color recognition concepts as previous sections. Added onto this, however, was the addition of an algorithm that would recursively go through board values according to whether they would result in a win or loss in the game of tic-tac-toe.

When no movement in frame was detected, the robot would look for whether or not there had been any changes in board states to determine if a new human move was made.

III. Results

Results obtained over the course of this lab were largely in line with those that were expected by the group and by the lab instructions. While there were initial programming complications with the robot's ability to smoothly implement real-time object tracking, these problems were eventually resolved by making changes to the robot's camera's masking that allowed images to be computed faster, in turn allowing for smoother motion of the robot when following a ball of a given color. Likewise, the robot's final ability to sort spheres by color was entirely successful. The only unresolved issue with the robot's behavior fell in the realm of its ability to play tic-tac-toe, as there were occasional instances of it misplacing balls outside of the established game board.

Following the group's completion of all necessary aspects of the final project, the group felt that the lab had done a good job of reenforcing all of the class's previous kinematics concepts while simultaneously applying them to an environment relevant and comparable to real-world devices. Robotic manipulators similar to or on a larger scale than those that we've used throughout RBE 3001 are extremely common in places like a factory setting, and in turn this lab provided a better understanding not only of the way in which this lab's robot's can function, but also the manner in which robots around the world can be utilized and programmed to perform a wide variety of different tasks.

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IV. Discussion

The lab had a rough start, as there was confusion with the camera giving unexpected errors. After a few attempts for the first sign offs, the problem was solved by adding white tape to the border of the checkerboard, as the camera was detecting extra spaces that did not exist on the checkerboard. The trouble did not stop there however, as the team divided to work on separate problems. Someone worked on sign off 2, 4, 5, and extra credit 3. Two team members worked on extra credit 3 extensively, however that task was unsuccessful, and was successfully replaced with extra credit 1. Another team member struggled with sign off two before getting assisted by the fourth member, who had also completed sign off 3, and four. The team had initially planned to play chess, but due to unforeseen delays, ended up coding tic-tac-toe as a bonus project instead.

A lot of time was lost due to difficulties troubleshooting sign off 1, even after asking the TAs, the error persisted before the group realized that it was an issue of the checkerboard alignment and lighting. After the camera started working, masking was applied, which removed the checkerboard background and converted it to black. Then the colored balls were not masked, and stood out against a black background, which then got additional color masks, to separate the red, orange, yellow, and green colors. One of the hardest challenges was doing the sign offs out of order. Sign off 5 was done with arbitrary points, while sign off 2 was still incomplete. Extra credit had been worked on after sign off 1 was completed, which was not ideal. Aside from the challenges of not completing the lab in the intended order, we struggled due to exhaustion thinking of how to complete sign off 4 for a while, before discussing with a physics associate who recommended a geometric approach. During this lab the group faced exhaustion and occupied schedules with finals weeks. There was a roadblock for a while with extra credit 1 trying to optimize the tracking of the arm, however rewriting the masking function efficiently made the arm run fast and in real time. The group finished the main project exhausted. However, this did not deter from the final goal of making the arm play a game. After a late night until 2:30 AM, and finals week sustenance provided by the department, as well as some time before lab period on demo day, tic tac toe was functional.

V. Conclusion

Over the course of the final project, the team was required to utilize a large amount of the skills and programming that were developed over the course of previous labs. This began with the use of forward and inverse kinematics, both with regard to position kinematics and velocity kinematics. In addition to these abilities from previous labs came practice with and exposure to camera calibration, concepts such as color masking for the sake of isolating different spaces and specific shapes within them, and utilizing these concepts alongside each other and the previous labs' topics in order to perform different types of automated tasks. Altogether, the lab served as a method of bringing together all of the concepts that were introduced and practiced over the course of RBE 3001.

VI. Code Release

https://github.com/RBE3001-C25/RBE3001_C25_Team_2/releases/tag/TicTacToe

VII. Team Video

https://www.youtube.com/watch?v=zCdq1DwnjjE

VIII. Team Member Controbutions

<table>
<tr><td>Planning</td><td>Primary Contribution</td><td>Secondary Contribution</td><td>Tertiary Contribution</td><td>Quaternary Contribution</td></tr>
<tr><td>Coding</td><td>Graham Tic-tac-toe, #2, #3, #4</td><td>Vishesh EC-1, #5, #6 (EC-3 not working</td><td>Olivia EC-1, #1, #3 (masking), #5 (EC-3 not working)</td><td>Henry #2</td></tr>
<tr><td>Experimenting</td><td>Graham</td><td>Vishesh/Olivia</td><td colspan="2">Henry</td></tr>
<tr><td colspan="2">Analyzing results</td><td colspan="3">Everyone</td></tr>
<tr><td>Report</td><td>Henry</td><td>Vishesh</td><td>Olivia</td><td>Graham</td></tr>
<tr><td>Video-</td><td colspan="2">Henry</td><td colspan="2">Vishesh/ Olivia/Graham</td></tr>
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Creation		